NewsBot Intelligence System

ITAI 2373 | NewsBot Midterm | Technical Research Report

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Project repository: https://github.com/jtuozzo1616/Mid-Term-Jose-Group-NewsBot-Intelligence-System

The results in this report come from the completed BBC experiment. Model settings, sample counts and source details are recorded in the execution manifest.

Executive summary

NewsBot is a small, interpretable, English-language newsroom-intelligence prototype, not an automated fact-checker. It combines eight NLP course modules in an end-to-end workflow: acquisition and business context, preprocessing, TF-IDF, POS patterns, syntactic relationships, sentiment and emotion proxies, supervised multi-class classification, and named entity recognition. It is designed to support human editorial routing, entity monitoring and cautious interpretation of historical article coverage.

Business case and audience

Editors need to triage articles quickly; media analysts need category-dependent terminology, sentiment context and named-entity summaries. Proposed operational metrics are topic-routing macro-F1, per-desk recall, review volume and frequency of entity extraction errors. The classifier does not validate whether reported statements are true. Manual review is essential for ambiguous predictions, names and consequential use cases.

Dataset provenance and compliance

The recommended source is Kaggle BBC News Classification, competition file BBC News Train.csv, with 1,490 labeled records before quality filtering (ArticleId, Text/Article, Category). Its five topical labels are business, entertainment, politics, sport, tech. It does not contain a Health label, despite that example appearing among the instructor’s illustrative system categories. The unlabeled competition test file must not be used to compute test metrics. The dataset consists of older 2004–2005 BBC reporting, so current-event or publisher-generalization claims are unsupported.

Dataset URL: https://www.kaggle.com/competitions/learn-ai-bbc/data. The instructor requires approval before using any Kaggle dataset. Full copyrighted source articles and API credentials must not be placed in a public repository.

Module 2 | Reproducible preprocessing

The loader validates schema, nonempty text, labeled classes, article length and sample counts. It removes exact duplicate source articles before splitting. Unicode whitespace, simple HTML markup, escaped entities, URLs and email artifacts are normalized while preserving names and quantitative information for grammar, entity and sentiment analysis. A separate lemmatized lexical view supports interpretation without stripping features from raw text needed for NER and syntactic analysis.

Module 3 | TF-IDF weighting and feature inspection

Descriptive corpus-level TF-IDF contrasts category mean weights with out-of-category mean weights for interpretable term charts. This exploratory analysis is explicitly separated from model evaluation: predictive vectorizers live inside scikit-learn Pipelines, so each cross-validation fold learns its vocabulary and IDF scores using training samples only. Parameters include feature cap, minimum and maximum document frequencies, sublinear term frequency and unigram/bigram alternatives.

Module 4 | Part-of-speech patterns

The spaCy small English pipeline labels grammatical tokens. NewsBot compares normalized document-level proportions of NOUN, PROPN, VERB, ADJ, ADV, ADP, NUM, PRON and AUX by category rather than raw totals, which would overcount longer documents. These comparisons are descriptive indicators of reporting style and topic, not causal descriptions of writing quality.

Module 5 | Dependencies, syntax and semantic approximation

Using spaCy dependency parses, the system records subject, object, noun chunk, sentence and passive structure counts. A limited rule extracts simple subject-verb-object tuples. These tuples are heuristics, not fully annotated semantic-role labels: they can miss passive voice, conjunctions, distant dependencies or implicit arguments. Linguistic annotations are capped at the first 3,500 characters per article for Google Colab Free CPU resource control; the classifier still receives full article text.

Module 6 | Sentiment and emotion proxies

NLTK VADER generates article-level continuous compound polarity and positive/neutral/negative labels using standard threshold values of ±0.05. An explanatory dictionary produces normalized lexical proxies for joy, fear, anger, sadness and surprise. The emotion proxies are not supervised or validated emotion classification, and polarity can reflect quotes or reported events rather than journalist approval. Comparisons aggregate the same metric by article category.

Module 7 | Supervised classification experimental protocol

After exact-text deduplication, the labeled corpus is stratified into 80% training and 20% untouched holdout, using random seed 42. A 5-fold stratified cross-validation experiment on the training subset compares Dummy majority, Multinomial Naive Bayes, Logistic Regression and LinearSVC with macro-F1 as the principal selection metric. GridSearchCV tunes only the best eligible non-dummy model using training folds, then the model is evaluated once on the holdout. Accuracy, macro- and weighted-F1, class-level precision/recall/F1, confusion matrix and error excerpts are exported for inspection.

Module 8 | Entity intelligence

spaCy extracts PERSON, ORG, GPE, DATE and MONEY from original-case text. The notebook reports target entity counts per article by category, prominent names and illustrative same-article PERSON/ORG co-mention links. A co-mention means only that two names appeared in the same analyzed text; it does not imply contractual, financial, political or causal relationships.

Integrated inference demonstration

The reusable NewsBotIntelligenceSystem class accepts title and body, applies the fitted classification pipeline, returns a category and a clearly labeled score type, and adds VADER polarity, dictionary emotion proxy, POS rates, approximate SVO tuples and named entities. Newly authored fictional technology, sports and politics articles test data not present in the supervised training corpus. Empty input raises a validation error. SVM softmax-transformed margins are explicitly uncalibrated and are not trustworthy probability estimates.

Measured results and error analysis

Final holdout results calculated from the completed notebook: Linear SVM; 5-fold tuned training CV macro-F1 0.9801, untouched holdout accuracy 0.9618, untouched holdout macro-F1 0.9605, evaluated on 1440 labeled, cleaned documents. This is within-corpus evaluation only.

Category	Precision	Recall	F1	Support
business	0.956	0.970	0.963	67
entertainment	0.980	0.925	0.951	53
politics	0.962	0.943	0.952	53
sport	0.958	1.000	0.978	68
tech	0.957	0.957	0.957	47

Baseline comparison and training-only feature ablation

Algorithm	CV macro-F1	CV accuracy
Linear SVM	0.9782	0.9791
Logistic Regression	0.9692	0.9705
Naive Bayes	0.9681	0.9696
Dummy majority	0.0769	0.2379

TRAIN-CV unigram macro-F1: 0.9776; unigram+bigram macro-F1: 0.9801. This comparison is descriptive about feature design and is not evaluated on the final holdout.

Selected model parameters: {'classifier__C': 2.0, 'tfidf__max_features': 10000, 'tfidf__ngram_range': [1, 2]}. Corpus source recorded as public_research_mirror_of_Kaggle_competition_CSV with SHA-256 f56a65dab2ea7285… (full fingerprint in manifest).

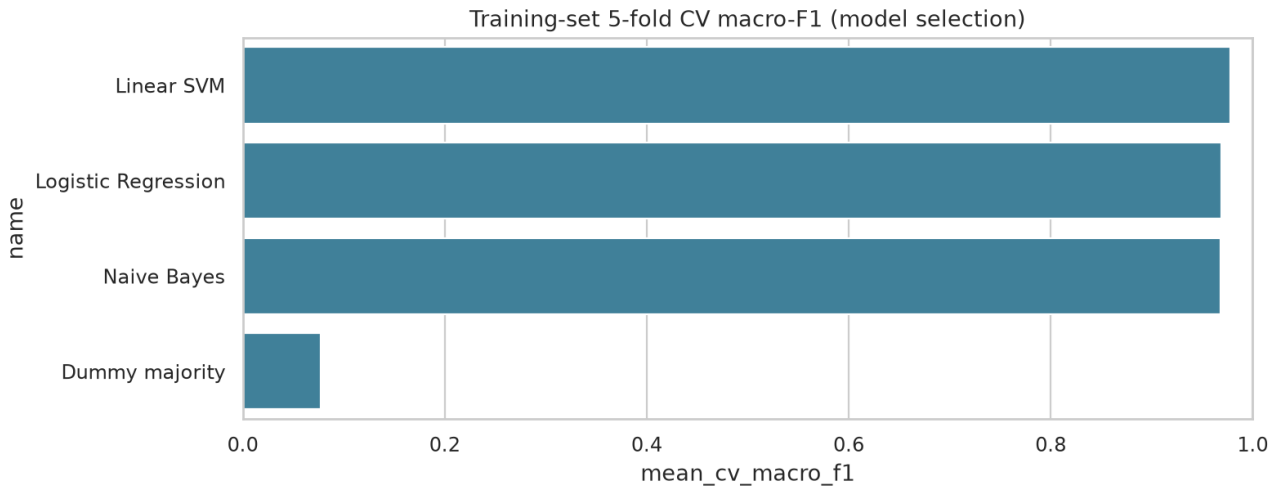
Diagnosis | What the 11 mistakes reveal

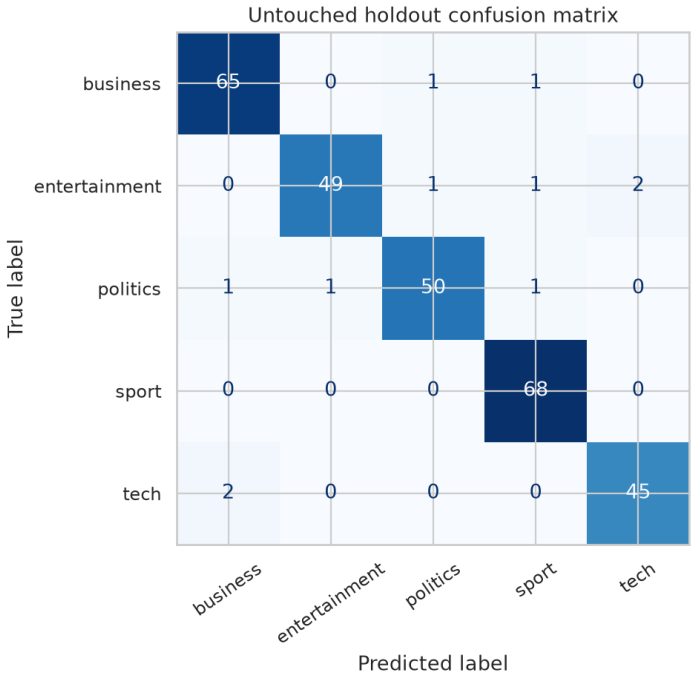
The holdout confusion matrix contains 11 misclassified articles (277/288 correct). Rather than treating each error as a random exception, the reviewed examples show potentially overlapping news desks: game-industry financial pressure, sports-club stock-market activity, televised sport and media-policy oversight. These are interpretations of observed topics, not verified explanations of the learned decision boundary.

Actual label	Predicted label	Errors
entertainment	tech	2
tech	business	2
business	politics	1
business	sport	1
entertainment	politics	1
entertainment	sport	1
politics	business	1
politics	entertainment	1
politics	sport	1

The most frequent directions are tech to business (2) and entertainment to tech (2); seven other directions occur once. In the entertainment class, 4 of 53 real articles were missed (recall 92.45%). Every one of the 68 actual sport articles was found (recall 100%), but three non-sport articles were incorrectly sent to sport, reducing sport precision to 95.77%.

Editorial implication: compare false deliveries and missed stories by desk, and send ambiguous cross-topic cases for human review. The model's SVM margins have not been calibrated as probabilities, so any automated threshold should be defined using new validation data, not tuned on the holdout. Full, copyright-safe error breakdown: NewsBot error case study on GitHub.

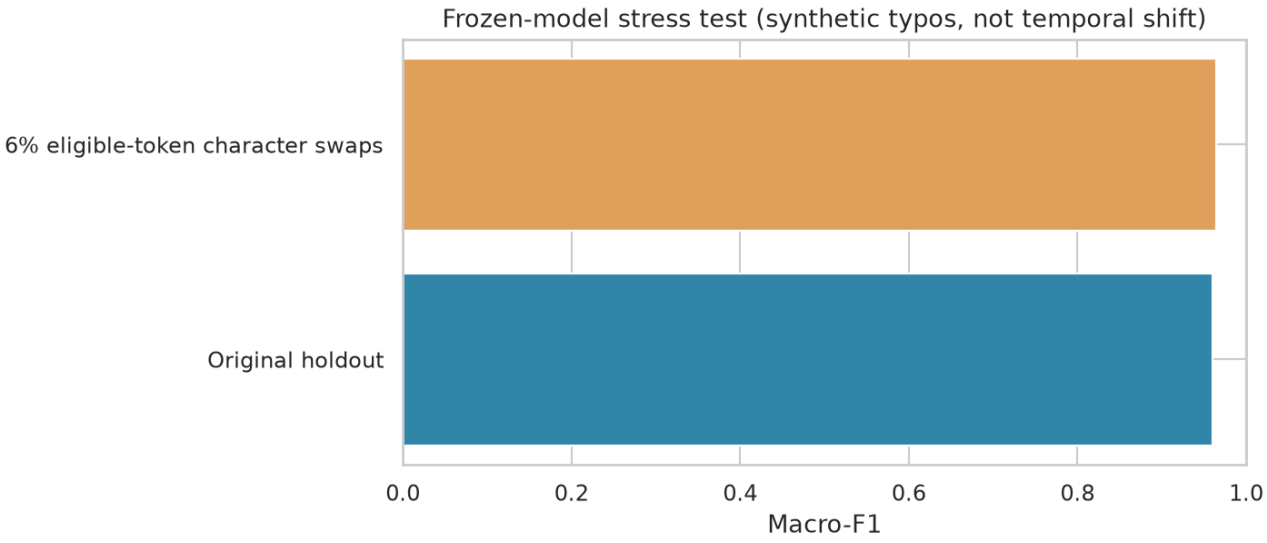




Research extension | Frozen-model robustness

Unmodified holdout macro-F1 0.9605; deterministic character-swap stress macro-F1 0.9639. This measures sensitivity to synthetic typos, not robustness to current reporting or independent publishers.

800-resample bootstrap 95% interval for historical holdout macro-F1: [0.9334, 0.9800]. This interval ignores systematic publisher and time shift.



Cross-module insights and business implications

VADER was lowest for politics (mean 0.054) and highest for entertainment (mean 0.590). This describes polarity of one historical source, not journalist intent or population emotion.

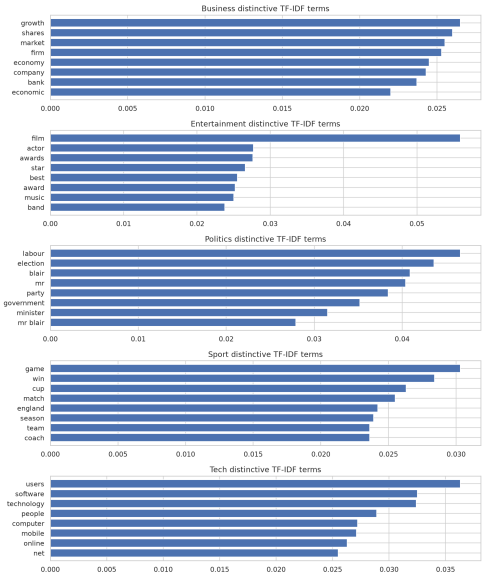
Category-distinctive TF-IDF expressions: business: growth, shares, market; entertainment: film, actor, awards; politics: labour, election, blair; sport: game, win, cup; tech: users, software, technology. Distinctive expressions support editorial keyword audits but are not proof of outside-domain generalization.

Entity mentions counted in annotated (length-capped) documents: DATE=8274, GPE=5222, MONEY=1667, ORG=4149, PERSON=7482. The totals may be distorted by story length, truncation and NER mislabels.

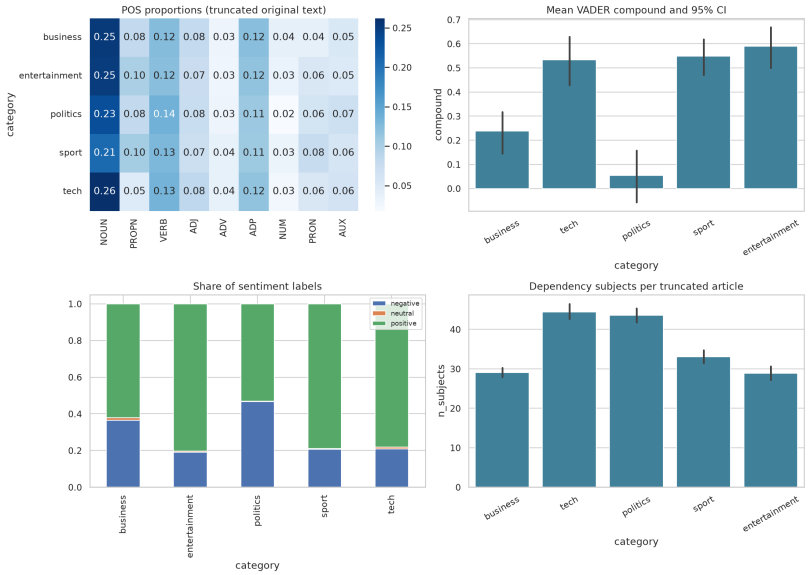
Descriptive POS contrast: the proportion of proper-noun tags (PROPN) was lowest for tech (0.050) and highest for sport (0.105). Linguistic annotations cover a limited text prefix and can be affected by source lowercasing.

The confusion directions are analyzed above using a count-only audit. Original news excerpts remain restricted to the execution evidence archive; only aggregate labels and careful topical paraphrases are published.

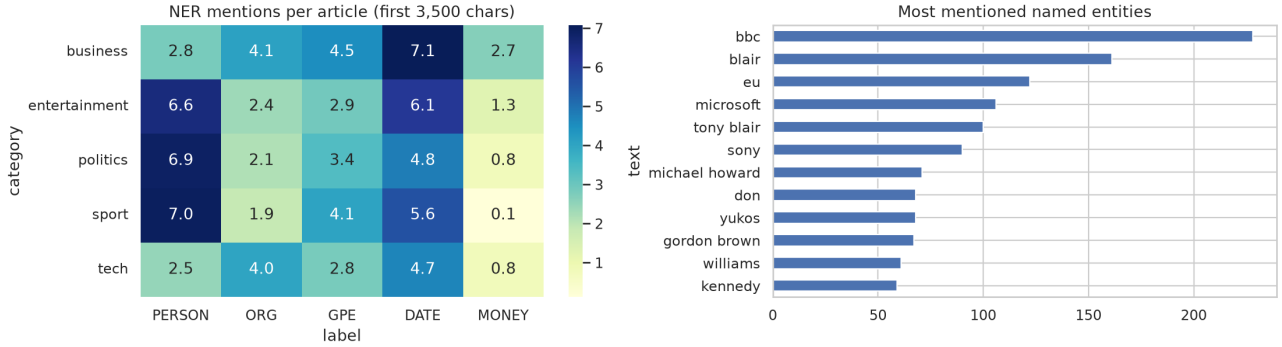
Distinctive historical category terms



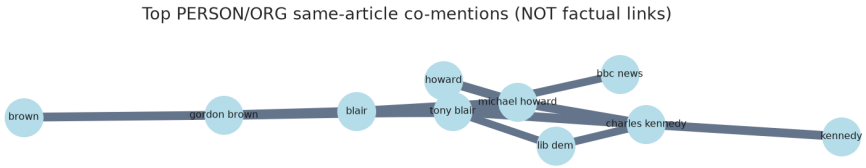
POS and sentiment patterns by topic



Named entity counts and frequency



Illustrative co-mention graph; not factual links





Risks, next steps and responsible deployment

This study uses single-source historical news and in-domain test splits. It does not establish robustness to 2026 articles, unseen publishers, healthcare news, multilingual input, implicit sentiment, factual truth or real-world decision thresholds. For a safer pilot: (1) require human review for unusual language or low relative model scores; (2) add external dataset evaluation; (3) check entity alias resolution; (4) monitor false categorization and distribution drift; (5) calibrate probabilities before using numeric review cutoffs; and (6) maintain versioned audit logs without unnecessarily storing personal data.

Project materials and submission

The public repository contains the complete eight-module notebook, model evaluation tables, visualizations, software tests and reproducibility instructions. A two-page reflective journal accompanies this report. The original assignment specifies team participation and requires approval for a Kaggle dataset, so the final submission format and dataset authorization should be confirmed with the instructor.

References

[1] BBC News Classification, Kaggle, https://www.kaggle.com/competitions/learn-ai-bbc/data. [2] Greene, D., and Cunningham, P., Practical Solutions to the Problem of Diagonal Dominance in Kernel Document Clustering, ICML (2006), source: https://github.com/derekgreene/bbc-datasets. [3] scikit-learn, Common pitfalls and data leakage: https://scikit-learn.org/stable/common_pitfalls.html. [4] spaCy linguistic features: https://spacy.io/usage/linguistic-features. [5] C. Hutto and E. Gilbert (2014), VADER, documentation: https://github.com/cjhutto/vaderSentiment. [6] Instructor ITAI 2373 Midterm assignment and starter notebook, October 2026.